

Foundations of Audio Signal Processing

§5 Analog to Digital Conversion

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WINTER TERM

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From analog to digital signals

Goal: Make **real**, physical signals (i.e., those mathematically modeled by CT-signals) accessible to a computer.

This process is called **analog to digital conversion** (ADC) and involves the following two basic steps:

1. By **sampling**, a CT-signal $f : \mathbb{R} \rightarrow \mathbb{R}$ is converted into a DT-signal $f_s : \mathbb{Z} \rightarrow \mathbb{R}$.
2. By **quantization**, the (amplitude) range of $f_s : \mathbb{Z} \rightarrow \mathbb{R}$ is mapped to a discrete grid, resulting in a signal $f_Q : \mathbb{Z} \rightarrow \mathbb{Z}$.

Signals processed by computers are of **finite duration** and **finite amplitude range**. After quantization one hence typically has a signal

$$f_Q : [a : b] \rightarrow [c : d]$$

(where $[e : f] := \{e, e + 1, \dots, f\} \subset \mathbb{Z}$ as usual).

In this chapter we introduce both steps of sampling and quantization.

5.1 Sampling

In the following we summarize the basic definitions and notions.

- (1) The procedure to generate a DT-signal by taking the values of a CT-signal on a discrete set of points (in time) is referred to as **sampling**.
- (2) The values of the DT-signal are then called **samples**, the points are called **sample points**.
- (3) In the case that any two neighboring sample points have the same distance $T > 0$ (equidistant), one also speaks of **T -sampling**. Then a CT-signal $f : \mathbb{R} \rightarrow \mathbb{C}$ is transformed into the DT-signal $x : \mathbb{Z} \rightarrow \mathbb{C}$ with

$$x(n) := f(T \cdot n).$$

- (4) For T -sampling, the number of samples per time unit is $\frac{1}{T}$ which is referred to as **sampling rate**. The sampling rate referring to a time unit of 1 second is measured in **Hertz** or simply **Hz**.

Synthesis functions

The transition from a CT-signal $f : \mathbb{R} \rightarrow \mathbb{C}$ to a sampled DT-signal $x : \mathbb{Z} \rightarrow \mathbb{C}$ results, in general, in a loss of information, i.e., f cannot be reconstructed from x .

At least, one can try to reconstruct an approximation of f . This is done by using so-called [synthesis functions](#).

For example, in the case of T -sampling one can choose the characteristic functions $\mathbf{1}_{[t_k, t_{k+1})}(t)$ of the intervals $[t_k, t_{k+1})$, $t_k := T \cdot k$, between two neighboring sample points.

Then one obtains the function f_T defined by

$$f_T(t) := \sum_{k=-\infty}^{\infty} x(k) \mathbf{1}_{[t_k, t_{k+1})}(t) = \sum_{k=-\infty}^{\infty} f(T \cdot k) \mathbf{1}_{[t_k, t_{k+1})}(t).$$

In this case f is approximated by f_T , i.e., the value $f(t)$ is approximated by the constant $f(t_k)$ where t_k is the closest sampling point to the left of t .

Synthesis functions (2)

Of course, synthesis functions are not uniquely determined. For example, instead of the characteristic functions one could also pick spline functions or trigonometric functions.

Hence, the quality of the approximation depends not only on the number and quality of the sample values but also on the choice of the synthesis function. However, there is no canonical choice. The “right” choice depends very much on the class of CT-signals under consideration and the application in mind.

For a certain class of CT-signals the choice of the so-called [sinc-functions](#) leads to a [perfect](#) reconstruction of the CT-signal from its samples. This amazing property is the content of the next section.

Shannon Sampling Theorem

The Fourier transform $\hat{f}$ of a continuous-time signal $f \in L^1(\mathbb{R})$ exhibits the spectral information on the signal. Intuitively, the value $|\hat{f}(\omega)|$ expresses the (average) intensity in which the frequency $\omega \in \mathbb{R}$ is “contained” in the signal f .

Sometimes it suffices to consider signals which only contain a certain range of frequencies.

For example, the human auditory system only recognizes frequencies which are below a certain threshold (around 20 kHz). Therefore, for a given audio signal wiping out all frequencies above this threshold results in a signal with no audible loss — at least for the human ear.

(A bat, however, would probably protest against this statement. . .)

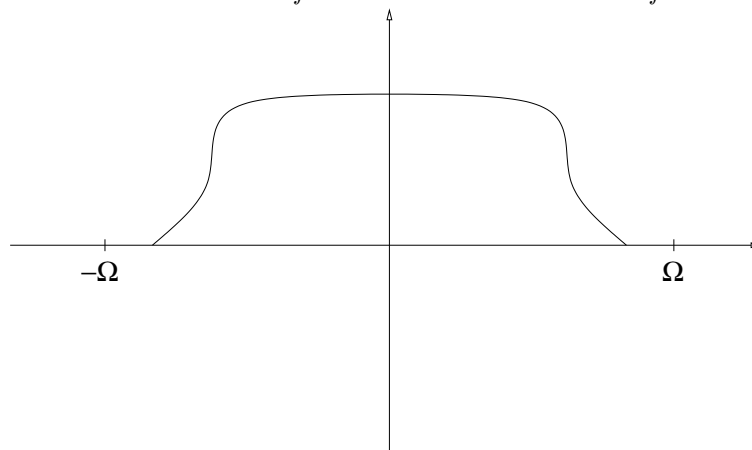
Bandlimited functions

Signals which contain only frequencies up to a certain threshold are called **bandlimited**:

Definition. Let $\Omega > 0$. The CT-signal $f \in L^2(\mathbb{R})$ is called **Ω -bandlimited** if the Fourier transform $\hat{f}$ vanishes for $|\omega| > \Omega$:

$$\hat{f}(\omega) = 0 \quad \forall |\omega| > \Omega, \quad \text{i.e., } \text{supp} \hat{f} \subset [-\Omega, \Omega].$$

The following figure shows a Fourier transform $\hat{f}$ of an Ω -bandlimited function f .



The Sampling Theorem

The sampling theorem by Shannon says that a bandlimited function f can be perfectly reconstructed from a suitable set of equidistant samples.

Theorem. [Shannon]¹ Let $f \in L^2(\mathbb{R})$ be an Ω -bandlimited function and let x be the T -sampled version of f with $T := \frac{1}{2\Omega}$, i.e., $x(n) = f(nT)$, $n \in \mathbb{Z}$.

Then f can be reconstructed from x :

$$f(t) = \sum_{n=-\infty}^{\infty} x(n) \operatorname{sinc}\left(\frac{t - nT}{T}\right) = \sum_{n=-\infty}^{\infty} f\left(\frac{n}{2\Omega}\right) \operatorname{sinc}(2\Omega t - n),$$

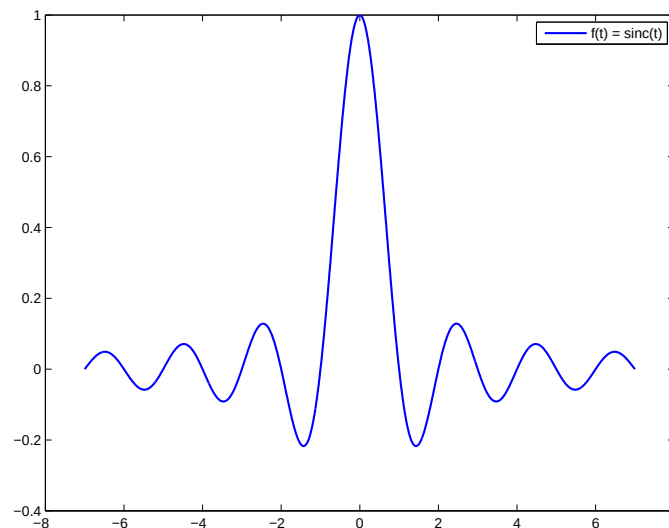
where the scaled and translated versions of the sinc-function

$$\operatorname{sinc}(x) := \begin{cases} \frac{\sin \pi x}{\pi x} & x \neq 0 \\ 1 & x = 0. \end{cases}$$

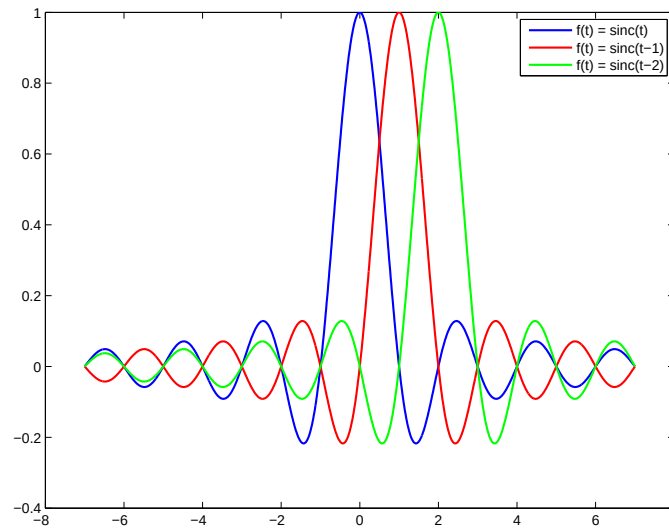
are used as synthesis function.

¹ Shannon(1949), Kotel'nikov(1933), Whittaker(1915), de la Vallée Poissin(1908)

The sinc function



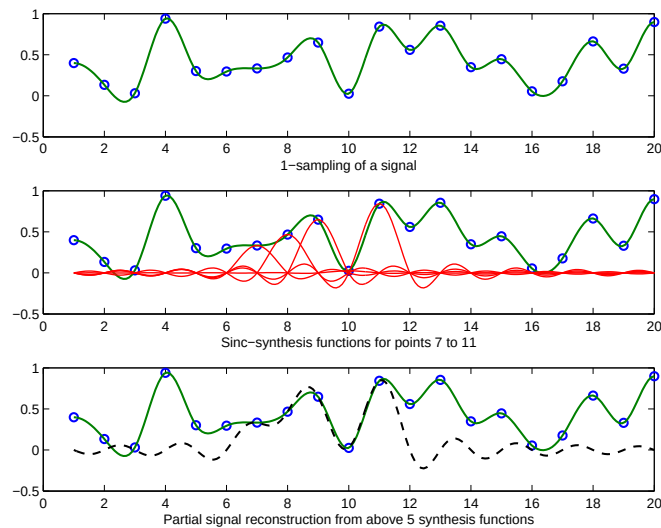
Shifted sinc functions



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Partial signal reconstruction using sinc-functions



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The Nyquist-Rate

Given an Ω -bandlimited function one needs a T -sampling with $T \leq \frac{1}{2\Omega}$. The sampling rate $\frac{1}{T} = 2\Omega$ Hz is sufficient for a perfect reconstruction of the signal.

Definition. For an Ω -bandlimited function, the sampling rate 2Ω is called **Nyquist-Rate**. Correspondingly, $\Omega = \frac{1}{2T}$ is called **Nyquist frequency** for the sampling interval of length T .

Example. Common sampling rates are:

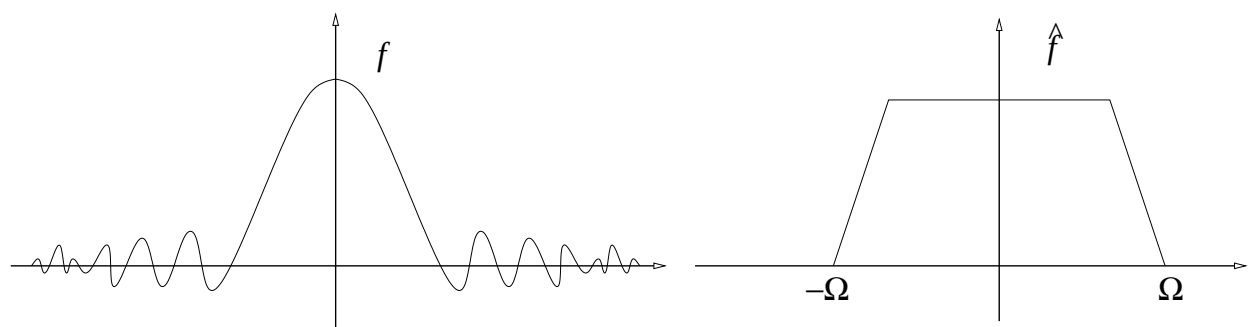
Device	sampling rate	reproducible frequency
Telephone	8 kHz	≤ 4 kHz
DAT	32 kHz	≤ 16 kHz
	44,1 kHz	$\leq 22,05$ kHz
	48 kHz	≤ 24 kHz
CD	44,1 kHz	$\leq 22,05$ kHz

As mentioned before, the human auditory system only recognizes frequencies up to 20 kHz. Thus the reconstructed digitized CD-signals sounds like the original music piece — there is no audible loss of information.

Aliasing

In the following, we illustrate what happens with the spectrum when sampling a signal.

Let $f \in L^2(\mathbb{R})$ be an Ω -bandlimited function as shown in the following figure.



Aliasing as periodization

Let $x : \mathbb{Z} \rightarrow \mathbb{C}$ be the DT-signal by sampling f with a sampling rate $\frac{1}{T}$, i.e.,

$$x(n) = f(nT), \quad n \in \mathbb{Z}.$$

Then the spectrum $\hat{x}$ of x is the 1-periodization of a scaled version of the spectrum $\hat{f}$.

The precise statement is:

Theorem. Let f be a continuous function with $f \in L^2(\mathbb{R})$ and x be the DT-signal obtained from f by T -sampling, i.e., $x(n) = f(nT)$ for $n \in \mathbb{N}$. Then

$$\hat{x}(\omega) = \frac{1}{T} \sum_{k \in \mathbb{Z}} \hat{f}\left(\frac{\omega + k}{T}\right).$$

Proof

We first prove the result for the case $T = 1$.

For $\omega \in \mathbb{R}$ let

$$g(\omega) := \sum_{k \in \mathbb{Z}} \hat{f}(\omega + k).$$

To show that $\hat{x} = g$ it suffices to show that their inverse Fourier transforms coincide, i.e., $x = \check{g}$. To this end, we compute the n th Fourier coefficient of g :

$$\begin{aligned} \int_0^1 g(\omega) e^{2\pi i \omega n} d\omega &= \int_0^1 \sum_{k \in \mathbb{Z}} \hat{f}(\omega + k) e^{2\pi i \omega n} d\omega \\ &= \sum_{k \in \mathbb{Z}} \int_k^{k+1} \hat{f}(\omega) e^{2\pi i (\omega - k)n} d\omega \\ &= \int_{\mathbb{R}} \hat{f}(\omega) e^{2\pi i \omega n} d\omega \\ &= f(n) = x(n). \end{aligned}$$

For general $T > 0$, we consider the function h defined by $h(t) := f(t \cdot T)$ instead of f in the above proof and the result follows from the properties of the CT Fourier transform.

Sampling at the Nyquist rate

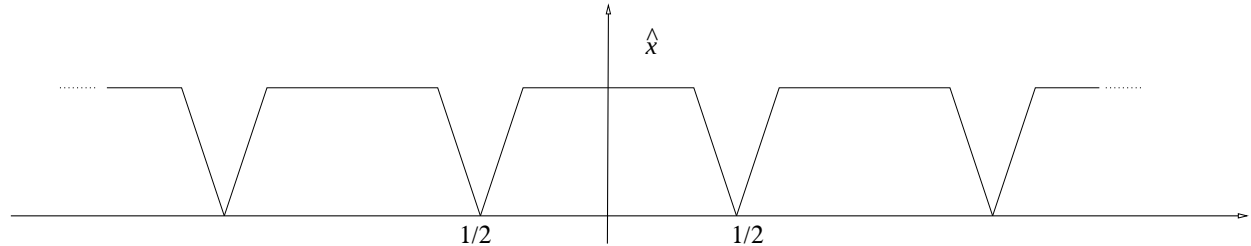
We now consider the three cases where the sampling rate $\frac{1}{T}$ is equal to, above and below the Nyquist rate 2Ω .

Case 1: $T = \frac{1}{2\Omega}$

When the sampling rate equals the Nyquist rate, the Fourier transform of $\hat{f}$ can be recovered from $\hat{x}$ since

$$\hat{f}(\omega) = \chi_{[-1/2T, 1/2T]} \cdot T \cdot \hat{x}(T\omega) = \chi_{[-\Omega, \Omega]} \cdot T \cdot \hat{x}(T\omega)$$

Therefore, f can be perfectly recovered from x by taking the inverse Fourier transform of $\chi_{[-\Omega, \Omega]} \cdot T \cdot \hat{x}(T\cdot)$. Actually, this is just a reformulation of Shannon's sampling theorem.

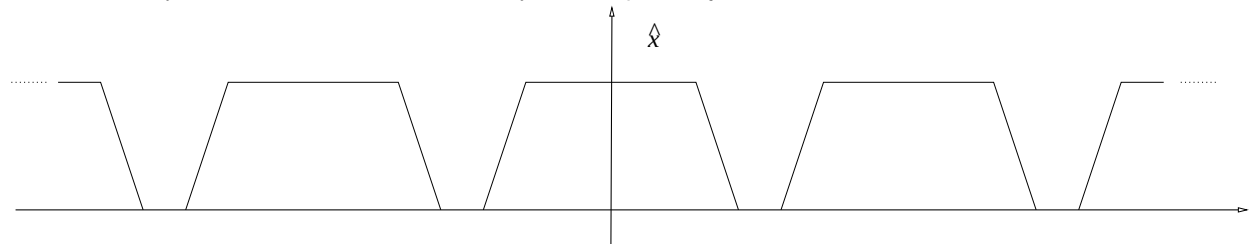


Oversampling

Case 2: $T < \frac{1}{2\Omega}$

In this case the sampling rate $\frac{1}{T}$ is above the Nyquist rate 2Ω . One then also speaks of **oversampling**.

As in Case 1, $\hat{f}$ can be recovered from $\hat{x}$ and f can be perfectly reconstructed from x .



One can show that in the case of oversampling, increasing the sampling rate leads to faster convergence properties of the reconstruction series $f(t) = \sum_{n=-\infty}^{\infty} x(n) \text{sinc}\left(\frac{1}{T}(t - nT)\right)$.

Undersampling

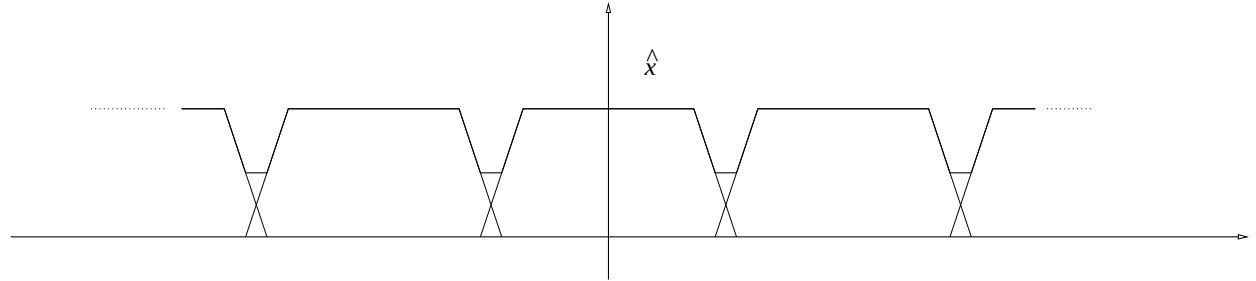
Case 3: $T > \frac{1}{2\Omega}$

In this case the sampling rate $\frac{1}{T}$ is below the Nyquist rate 2Ω . One then also speaks of **undersampling**.

In this case, reconstruction of f from x is not any longer possible.

In the spectrum of $\hat{x}$ there is an “overlap” of the $\frac{1}{T}$ -translated spectra of $\hat{f}$ and $\hat{f} = \mathbf{1}_{[-\Omega, \Omega]} \hat{x}$ does not hold any longer.

This effect is known as **aliasing**.



Aliasing: mathematical analysis

We investigate the aliasing effect in a more rigorous way. Let $T = \frac{1}{2\Omega}$ be a sampling rate for some $\Omega > 0$ and $f \in L^2(\mathbb{R})$ only be an Ω' -bandlimited function for some Ω' with

$$\Omega < \Omega' < 3\Omega.$$

Then T -sampling of f leads to the undersampled DT-signal x with $x(k) := f(kT)$. By the Fourier inversion we obtain

$$\begin{aligned} f(kT) &= \int_{-\Omega'}^{\Omega'} \hat{f}(\omega) e^{2\pi i \omega k T} d\omega \quad (\text{as } \hat{f} = 0 \text{ for } |\omega| > \Omega') \\ &= \int_{-3\Omega}^{3\Omega} \hat{f}(\omega) e^{2\pi i \omega k T} d\omega \\ &= \int_{-3\Omega}^{-\Omega} \hat{f}(\omega) e^{2\pi i \omega k T} d\omega + \int_{-\Omega}^{\Omega} \hat{f}(\omega) e^{2\pi i \omega k T} d\omega + \int_{\Omega}^{3\Omega} \hat{f}(\omega) e^{2\pi i \omega k T} d\omega. \end{aligned}$$

Aliasing: mathematical analysis (2)

By a suitable substitution and using $e^{\pm 2\pi i 2\Omega kT} = 1$ one obtains

$$x(k) = f(kT) = \int_{-\Omega}^{\Omega} \underbrace{(\hat{f}(\omega) + \hat{f}(\omega + 2\Omega) + \hat{f}(\omega - 2\Omega))}_{=:(*)} e^{2\pi i \omega kT} d\omega \quad (1)$$

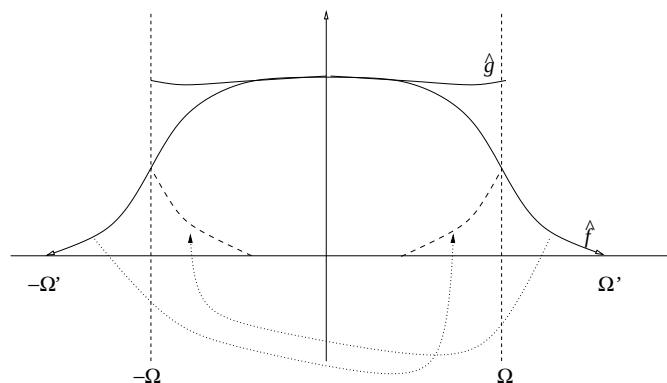
We define a function g via its Fourier transform:

$$\hat{g}(\omega) := \begin{cases} (*) & \text{if } |\omega| \leq \Omega \\ 0 & \text{otherwise.} \end{cases}$$

Then, g is Ω -bandlimited and by Equation (1) holds $g(kT) = f(kT)$ for all $k \in \mathbb{Z}$.

Conclusion: The reconstruction of the CT-signal from the undersampled DT-signal x (by means of the sinc-synthesis functions of the Shannon Theorem) does not result in the original signal f but in the signal g .

Aliasing as spectral folding



In $\hat{g}$, high frequencies above $|\Omega|$ are "folded" into the interval $[-\Omega, \Omega]$.

⇒ In the undersampled DT-signal x , high frequency components of f outside the interval $[-\Omega, \Omega]$ **cannot be distinguished** from certain low frequency components of f within $[-\Omega, \Omega]$, i.e., high-frequency components act as lower frequencies.

This effect is known as **aliasing**.

Sampling: concluding questions

Theorem. [Shannon] Let $f \in L^2(\mathbb{R})$ be an Ω -bandlimited function and let x be the T -sampled version of f with $T := \frac{1}{2\Omega}$, i.e., $x(n) = f(nT)$, $n \in \mathbb{Z}$.

Then f can be reconstructed from x :

$$f(t) = \sum_{n=-\infty}^{\infty} x(n) \operatorname{sinc}\left(\frac{t - nT}{T}\right) = \sum_{n=-\infty}^{\infty} f\left(\frac{n}{2\Omega}\right) \operatorname{sinc}(2\Omega t - n),$$

- Many "real" analog signals contain clicks and other discontinuities.
- Those are hence not bandlimited!
- How do we achieve that a signal f is Ω -bandlimited?
- $\rightarrow$ filters will be introduced in the next chapter!

5.2 Quantization

Basic principle: approximate real amplitudes by values from a finite codebook.

We partition the range $\mathbb{R}$ into intervals

$$\mathbb{R} = I_1 \sqcup \dots \sqcup I_N$$

where

$$I_j = (x_{j-1}, x_j], \quad y_j \in I_j$$

are suitably chosen.

We obtain a **codebook**

$$\{y_1, \dots, y_N\}$$

consisting of **codewords** y_j .

Quantization: Coder and decoder

The step of quantization is performed by the so-called **coder**

$$I_j \ni r \mapsto j.$$

The reverse step is carried out by a **decoder**

$$j \mapsto y_j.$$

Remark: In general, quantization is a **lossy** operation. I.e. the coder mapping is not invertible!

$R := \lceil \log_2 N \rceil$ is called the **bit rate** of the quantizer.

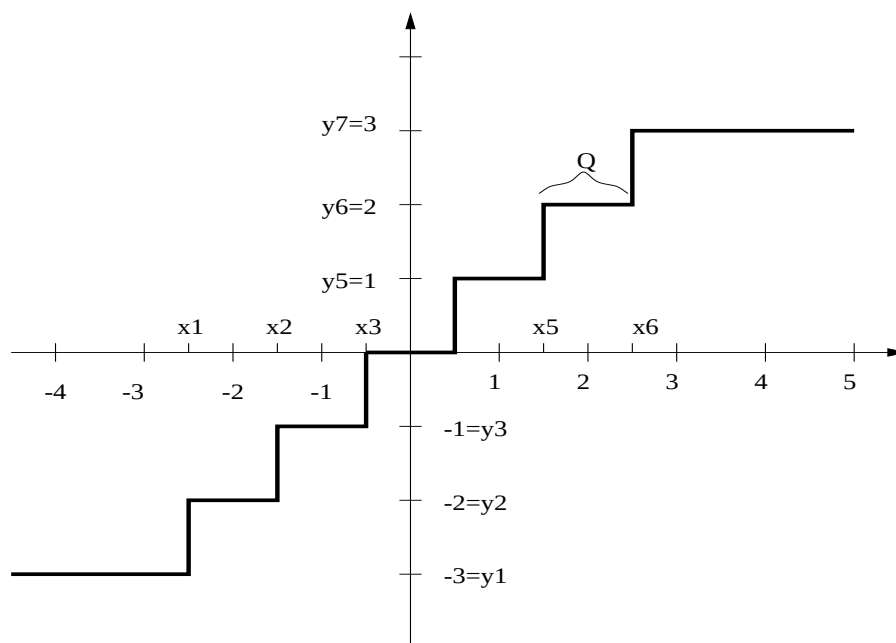
Example: A standard audio-CD has a bit rate of 16 bit using a uniform scalar quantizer.

Example: Uniform Scalar Quantizer

- Let $N = 7$.
- Consider a uniform width $Q = 1$ of the **quantizer levels**,
- **decision thresholds** $x_1, \dots, x_6$,
- codewords $y_1, \dots, y_7$ and
- a partition

$$I_1 = \left(-\infty, -\frac{5}{2} \right], I_2 = \left[-\frac{5}{2}, -\frac{3}{2} \right], I_3 = \left[-\frac{3}{2}, -\frac{1}{2} \right], I_4 = \left[-\frac{1}{2}, \frac{1}{2} \right], \\ I_5 = \left[\frac{1}{2}, \frac{3}{2} \right], I_6 = \left[\frac{3}{2}, \frac{5}{2} \right], I_7 = \left[\frac{5}{2}, \infty \right)$$

Graph of a Quantizer (Uniform Scalar Quantizer)



Quantization error

We again consider a uniform scalar quantizer with

- an amplitude range of $\pm x_{\max}$,
(samples of higher magnitudes are "clipped" to $\pm x_{\max}$)
- a bit rate of w ,
- and assume a uniformly distributed quantizer error. The corresponding probability density function (pdf) is denoted by f_E .

We obtain the following width of the quantizer levels:

$$Q = \frac{2x_{\max}}{2^w}.$$

The uniform pdf corresponding to Q is

$$f_E(x) := \frac{1}{Q} \delta_{|x| \leq Q/2}.$$

Signal-to-Noise Ratio

The resulting variance of the quantization error is

$$\sigma_E^2 = \int_{\mathbb{R}} x^2 f_E(x) dx = \frac{1}{Q} \left[\frac{x^3}{3} \right]_{-Q/2}^{Q/2} = \frac{Q^2}{12}.$$

Using the so-called *crest factor*

$$P_F := \frac{x_{\max}}{\sigma_X} = \frac{2^{w-1} \cdot Q}{\sigma_X}$$

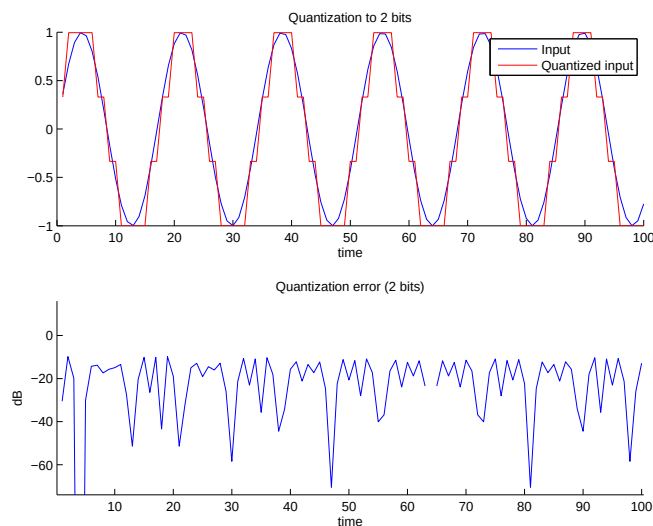
of the input signal x we obtain the **SNR (Signal-to-Noise Ratio)** as a measure of quality for our quantizer:

$$\begin{aligned} \text{SNR} &= \text{SNR}(X, E) := 10 \log_{10} \left(\frac{\sigma_X^2}{\sigma_E^2} \right) = 10 \log_{10} \left(\frac{3 \cdot 2^{2w}}{P_F^2} \right) \\ &= 6.02 \cdot w - 10 \log_{10}(P_F^2/3), \end{aligned}$$

→ the SNR is increased by approximately 6 dB for each bit of the bit rate w .

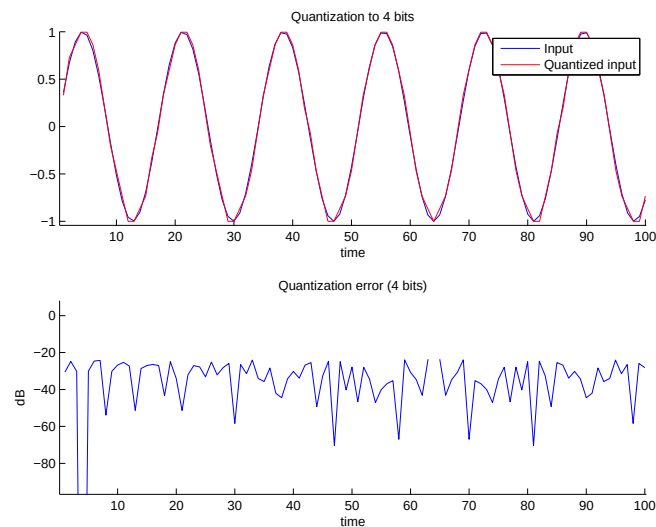
Example: Uniform Scalar Quantization

Sine wave quantized using $w = 2$ bits and reconstruction error in dB.



Example: Uniform Scalar Quantization

Sine wave quantized using $w = 4$ bits and reconstruction error in dB.



The (pseudo-) unit decibel (dB)

Remarks:

- For audio signals the unit decibel (short: dB) is usually used to measure the [sound pressure level](#).
- Strictly speaking, the dB is **dimensionless** as it measures the ratio (or difference) of two quantities (example: signal and noise).
- In case dB is used for measuring sound pressure level without specified reference quantity, the [threshold in quiet](#) of 20 micropascals ($2 \cdot 10^{-5}$ Pa) is assumed.

Typical sounds and sound pressure levels

Source	Sound pressure (Pa)	Sound pressure level (dB)
threshold in quiet (@ 1 kHz)	$2 \cdot 10^{-5}$	0
normal breathing	$6.3 \cdot 10^{-5}$	10
quiet room	$2 \cdot 10^{-4} - 6.3 \cdot 10^{-4}$	20-30
conversation	$2 \cdot 10^{-3} - 6.3 \cdot 10^{-3}$	40-50
TV (@ 1m)	0.02	60
car noise (@ 10 meters)	0.02-0.2	60-80
main road	0.2 - 0.63	80-90
disco/air hammer	2	100
threshold of pain	100	134

Summary: quantizer characteristics

- **range** $\pm x_{\max}$ $\sim$ of covered amplitude values
- **bit rate** $\sim$ resolution in the covered amplitude range
- **codebook** $\sim$ precise specification of the quantizer
(**graph** of the quantizer)

In our examples we only considered **uniform quantizers** (uniform distance between neighboring elements of the codebook).

In applications one frequently deals with **non-uniform quantisation** to improve resolution for particular parts of the amplitude range.

Examples:

- improved representation of silent parts of audio,
 - lossy audio coding (such as "mp3"/AAC/Dolby AC/...).
- (although, quantization is not performed on the time-domain signal!)

Non-uniform quantization

Non-uniform quantization may be performed using the following steps:

1. **Coder:**
 - (a) non-linear deformation of the amplitude range using a **compressor**-function
 - (b) uniform scalar quantization
2. **Decoder:**
 - (a) uniform reverse quantization
 - (b) inversion of the compressor function using a suitable **expander**-function

Remark: This approach is also called **companding** (compressor+expander))

The compressor/expander functions are sometimes called compressor/expander **law**.

Example: non-uniform quantization

A frequently used compressor function is the **μ -Law**, which yields a logarithmic quantization:

For a real $\mu > 0$, the μ -law compressor c_μ is defined by

$$c_\mu(x) := x_{\max} \frac{\ln\left(1 + \frac{\mu|x|}{x_{\max}}\right)}{\ln(1 + \mu)} \cdot \text{sign}(x).$$

Example: μ -law compressor function

